THE UK UNIVERSITY INTEGRATION BEE 2025/2026

$$\int$$

ROUND TWO

MARK SCHEME

Sponsored by

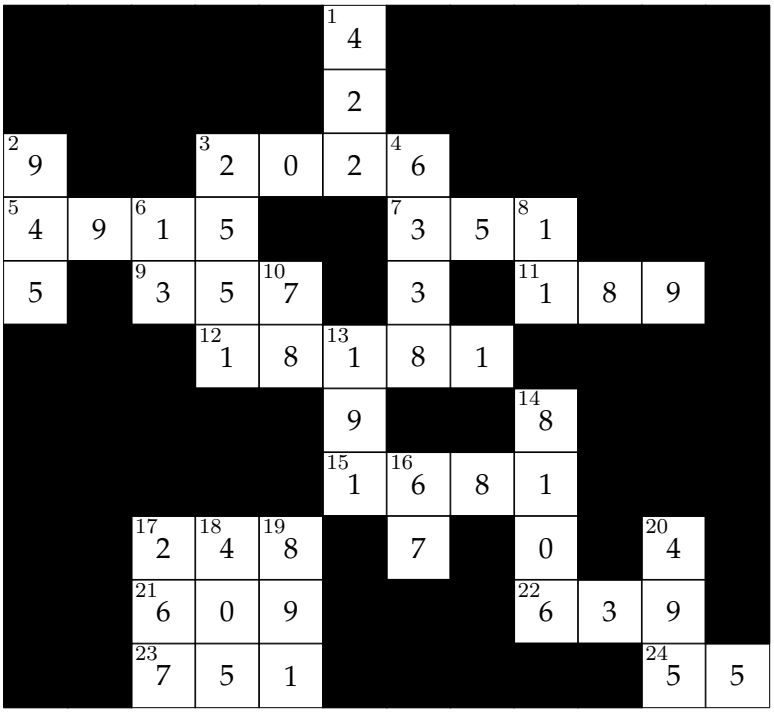


Jane Street

Group Round

1. 0
2. $\dfrac{\pi}{\sqrt{3}}$
3. $\dfrac{\pi}{2}$
4. $2 - \dfrac{\pi^2}{6}$
5. $\dfrac{\pi}{\sqrt{2}} - \dfrac{\pi}{4} - \dfrac{\ln(2)}{2}$
6. π
7. $\dfrac{\pi}{2e\sqrt{2}}$
8. π
9. $\ln(2) - 1$
10. 1

Crossnumber Round



Reminders

- Only one attempt per square.
- Once an attempt on a square is made, no matter if it's right or wrong, write the correct number in the master copy.
- Also put a small tick or cross in the master copy to indicate if the team got that square right.
- At the end, count the number of ticks in the master copy and write it down.

Relay Round

1. $-\dfrac{1}{2025}$
2. $\ln(2) - \dfrac{\pi}{4} + \dfrac{\pi^2}{32}$
3. $\begin{cases} \dfrac{2(n+2026)^2}{2n+53}, & m = n; \\ 0, & m \neq n. \end{cases}$
   (Reject partial answers.)
4. $\dfrac{\pi}{4}$
5. $\dfrac{(-1)^m m!}{n}\zeta(m+1)$
6. $\dfrac{1}{\varphi}\ln|1+x^{\varphi}| + C$
   (Absolute value not necessary. Normal brackets are fine.)
7. $1 - \dfrac{1}{\sqrt{2}}$
8. 0
9. $\dfrac{\pi}{4}$
10. $C \cdot 2025^x \cdot \Gamma\left(x + \dfrac{2026}{2025}\right)$
    $C \cdot 2025^x \cdot \left(x + \dfrac{1}{2025}\right) \cdot \Gamma\left(x + \dfrac{1}{2025}\right)$
11. $-\dfrac{\pi\gamma}{2}$
12. $\dfrac{1}{2}$
13. $\sqrt{\pi}$
14. $\dfrac{\pi}{e}$

15. $\frac{\pi}{3}$
16. $\frac{1}{e}\ln\left(\sqrt{2}+1\right)$
17. $\frac{\pi^2}{24}$
18. $\frac{\pi}{e}-\frac{\pi}{2}$
19. $\frac{5}{4}$
20. 1999

Reminders

- Answers must be in closed form. Reject infinite sums, infinite products, etc.
- C denotes an arbitrary constant. It is not necessary to have C in the case of indefinite problems.
- We have provided alternative answers where reasonably necessary, but the list is not exhaustive. Please use your best judgement.